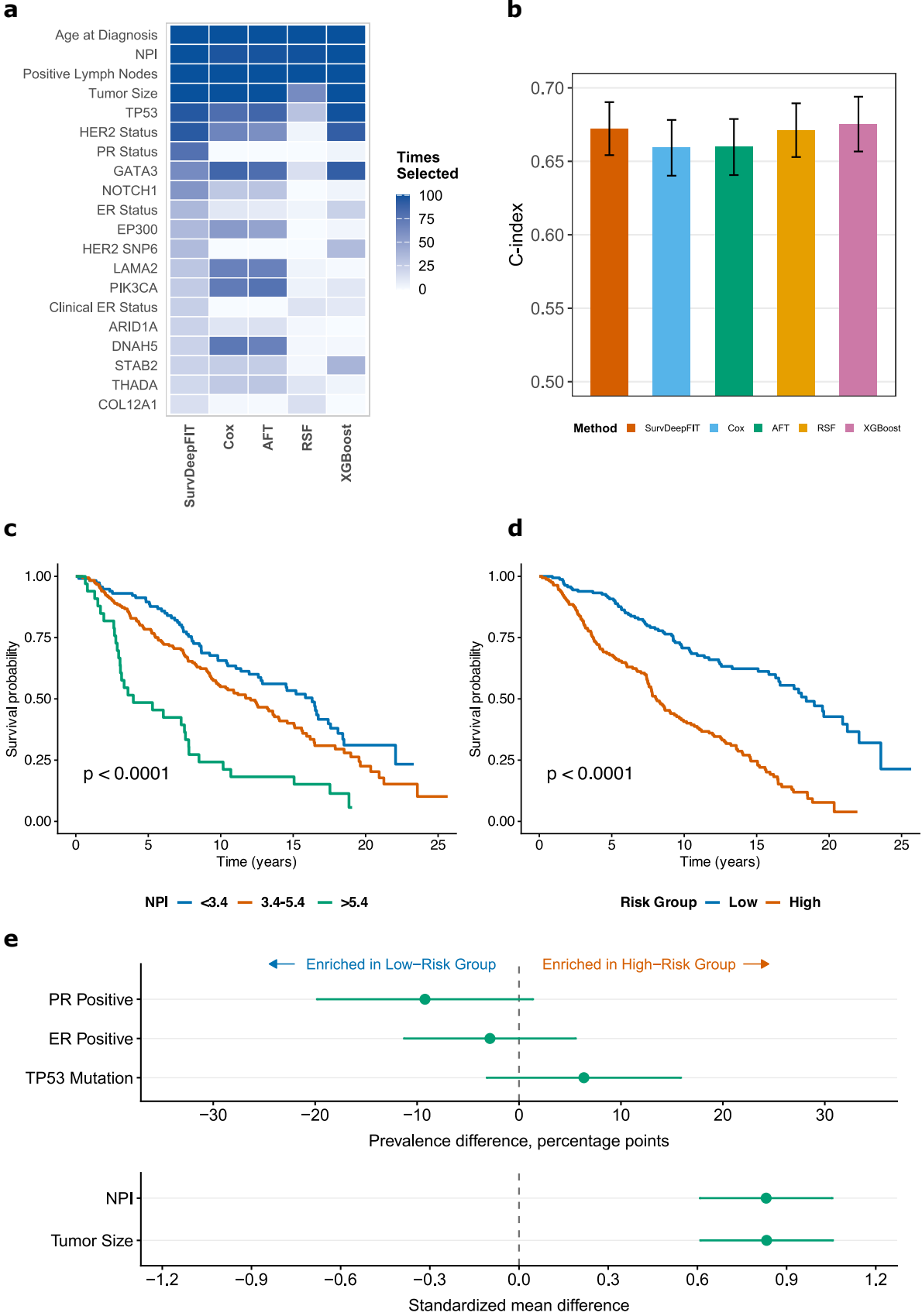

a
Age at Diagnosis
NPI
Positive Lymph Nodes
Tumor Size
TP53
HER2 Status
PR Status
GATA3
NOTCH1
ER Status
EP300
HER2 SNP6
LAMA2
PIK3CA
Clinical ER Status
ARID1A
DNAH5
STAB2
THADA
COL12A1
SurvDeepFIT
Cox
AFT
RSF
XGBoost
Times Selected
100
75
50
25
0
b
C-index
0.70
0.65
0.60
0.55
0.50
Method
SurvDeepFIT
Cox
AFT
RSF
XGBoost
c
Survival probability
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
5
10
15
20
25
Time (years)
NPI
<3.4
3.4-5.4
>5.4
d
Survival probability
1.00
0.75
0.50
0.25
0.00
p < 0.0001
0
5
10
15
20
25
Time (years)
Risk Group
Low
High
e
Enriched in Low−Risk Group
Enriched in High−Risk Group
PR Positive
ER Positive
TP53 Mutation
−30
−20
−10
0
10
20
30
Prevalence difference, percentage points
NPI
Tumor Size
−1.2
−0.9
−0.6
−0.3
0.0
0.3
0.6
0.9
1.2
Standardized mean difference